

Lecture 6: Induction

A simple statement

- Claim: Every number is even or odd.
- Even x : $\exists k, 2 * k = x$
- Odd x : $\exists k, 2 * k + 1 = x$

Naïve Approach

- $n = 5$: $\exists 2, 2*2 + 1 = 5$ so 5 is odd (so 5 is even or odd)
- $n = 9$: $2*4 + 1 = 9$ so 9 is odd
- $n = 14$: $2*7 = 14$ so 14 is even ($\rightarrow$ Even 14 $\vee$ Odd 14)

Methodical Approach

- $n = 0$. $2*0 = 0$ so n is even
- $n = 1$. $2*0 + 1 = 1$ so n is odd
- $n = 2$. $2*1 = 2$ so n is even
- $n = 3$. $2*1 + 1 = 3$ so n is odd
- ...

Induction

- For any statement P about n
- Base case: $P\ 0$ is true.
- Inductive case: $P\ k \rightarrow P\ (k+1)$
- THEN: P is true for all natural numbers n .

Proof of Even or Oddness

- Base case: $n = 0$.
- Claim: 0 is even or odd.
- Proof.
- $0 = 2 * 0$.
- Hence 0 is even.
- Hence 0 is even or odd.

Inductive case

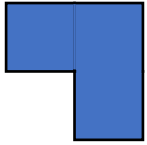
- Claim: $k+1$ is even or odd.
- Inductive hypothesis: k is even or odd.
- Consider k is even:
 - $2*j = k$.
 - $2*j + 1 = k + 1$
 - $\exists x, 2*x + 1 = k + 1$.
 - $k + 1$ is odd
 - Therefore $k + 1$ is even or odd.

Inductive case

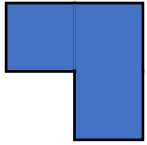
- Claim: $k+1$ is even or odd.
- Inductive hypothesis: k is even or odd.
- Consider k is odd:
 - $2*j + 1 = k$.
 - $2 * j + 2 = k + 1$
 - $2 * (j + 1) = k + 1$
 - $\exists x, 2 * x = k + 1$.
 - $k + 1$ is even
 - Therefore $k + 1$ is even or odd.

Triominoes

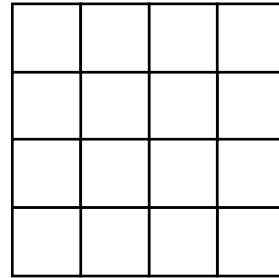
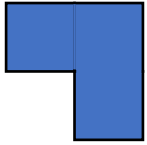
Triominoes



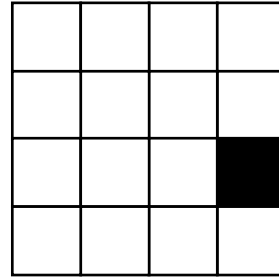
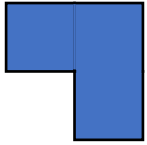
Triominoes



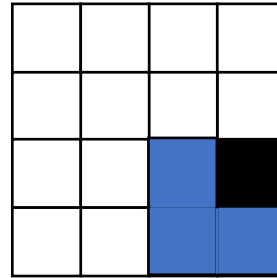
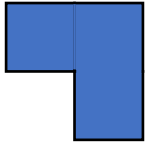
Triominoes



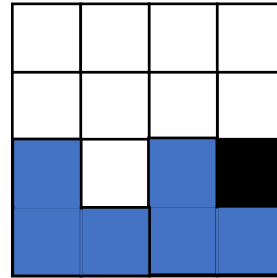
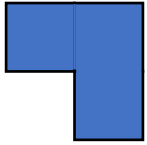
Triominoes



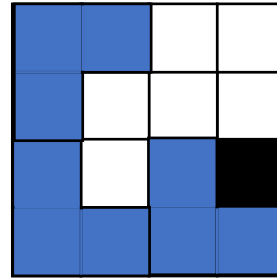
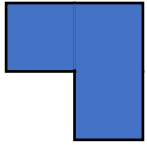
Triominoes



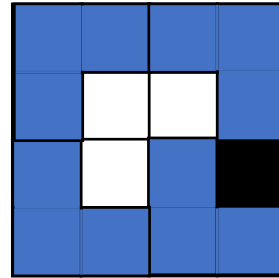
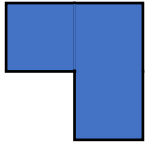
Triominoes



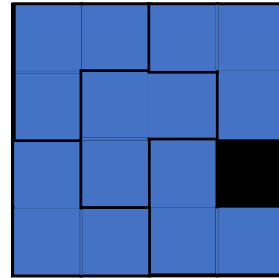
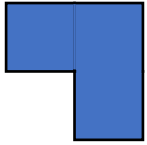
Triominoes



Triominoes

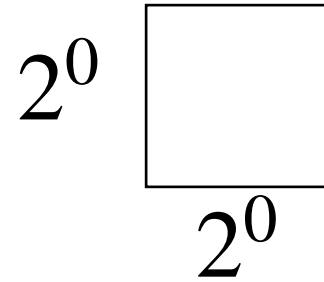


Triominoes



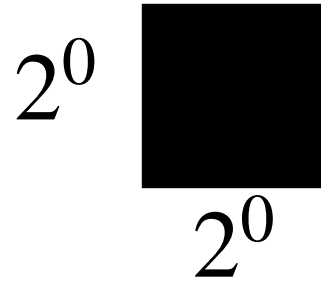
Triominoes: Base Case

$$n = 0$$



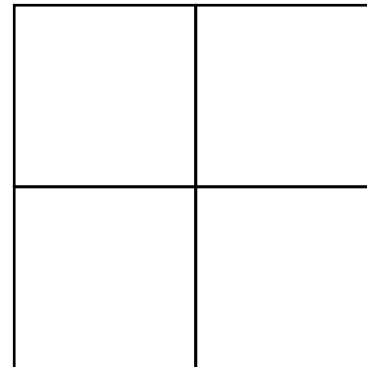
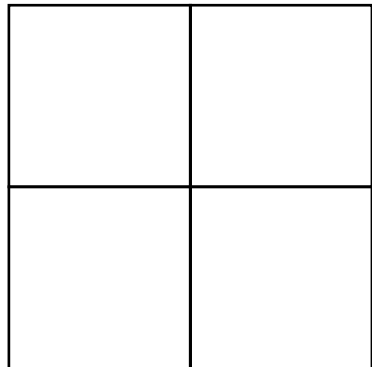
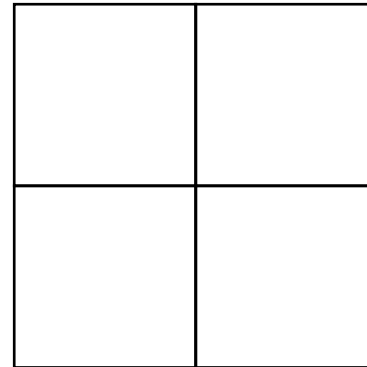
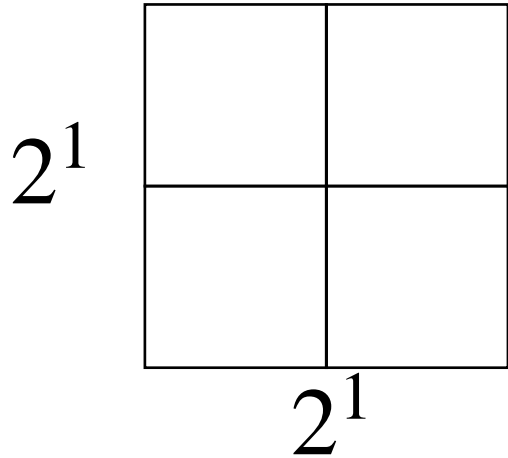
Triominoes: Base Case

$$n = 0$$



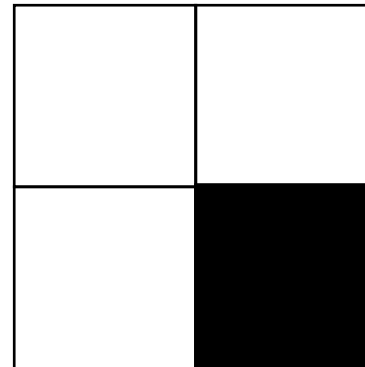
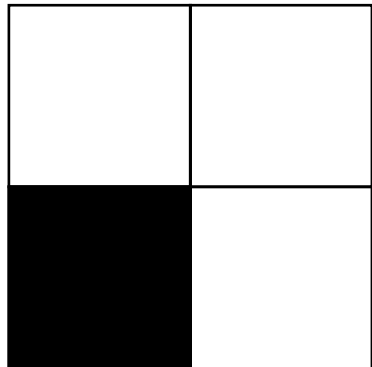
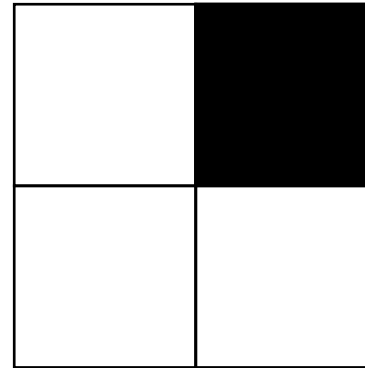
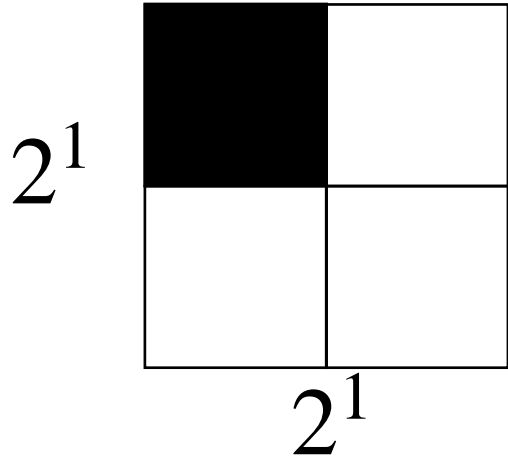
Triominoes: Sanity check case

$$n = 1$$



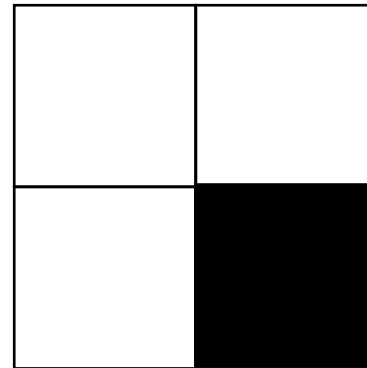
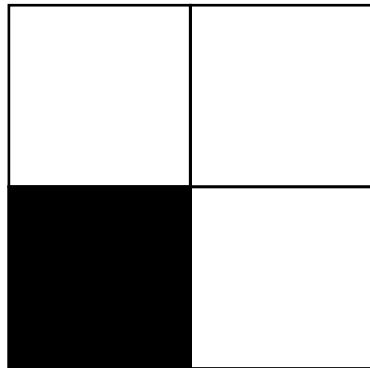
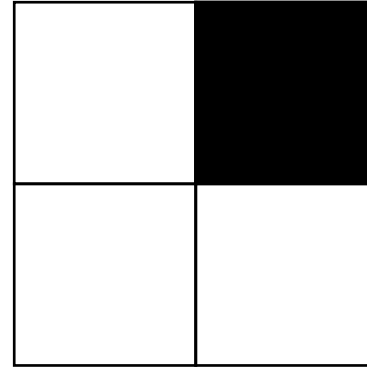
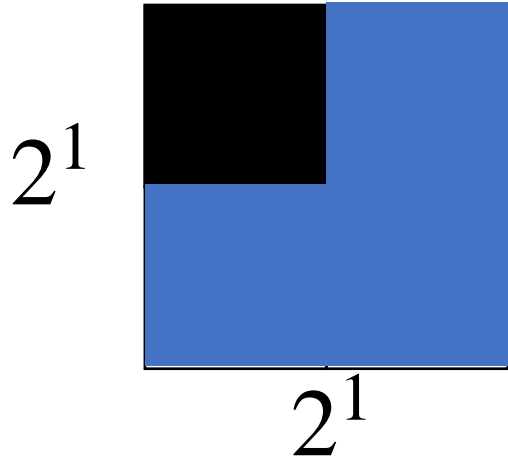
Triominoes: Sanity check case

$$n = 1$$



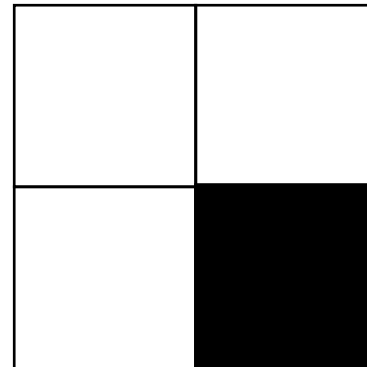
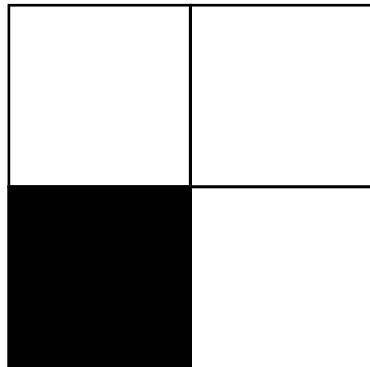
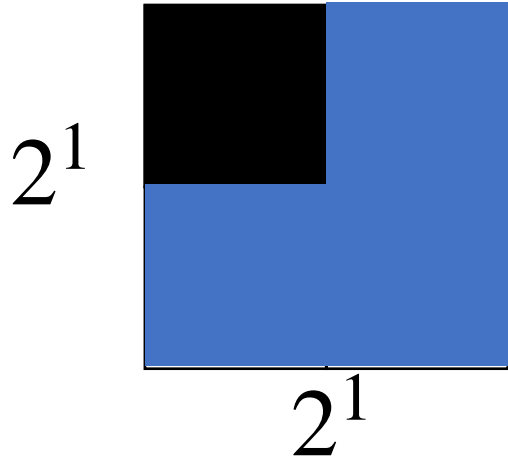
Triominoes: Sanity check case

$$n = 1$$



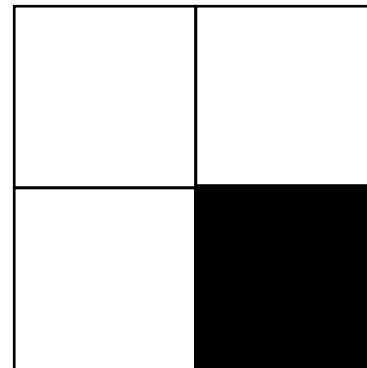
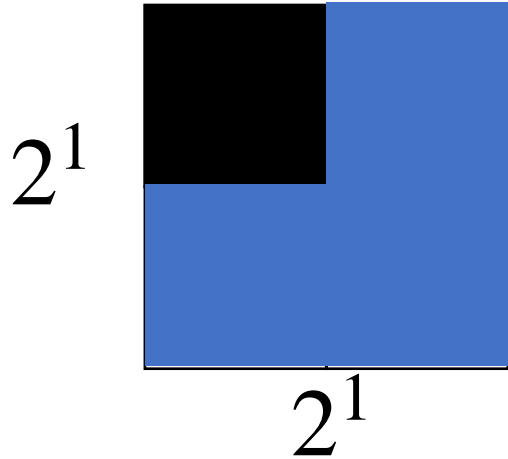
Triominoes: Sanity check case

$$n = 1$$



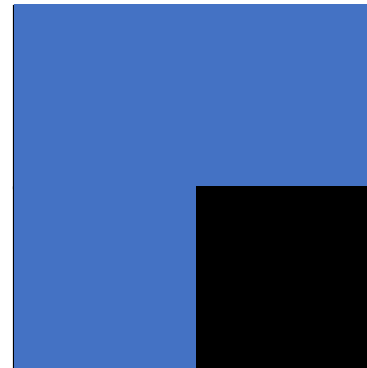
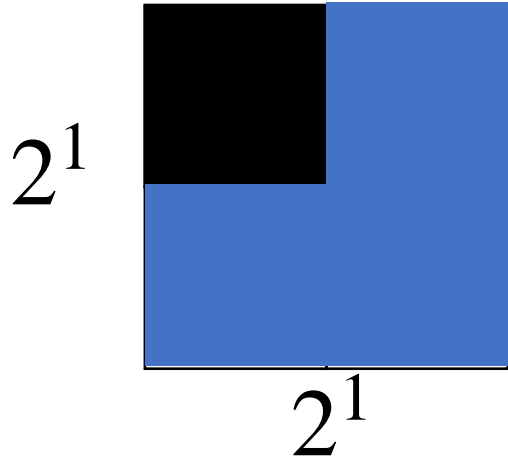
Triominoes: Sanity check case

$$n = 1$$



Triominoes: Sanity check case

$$n = 1$$

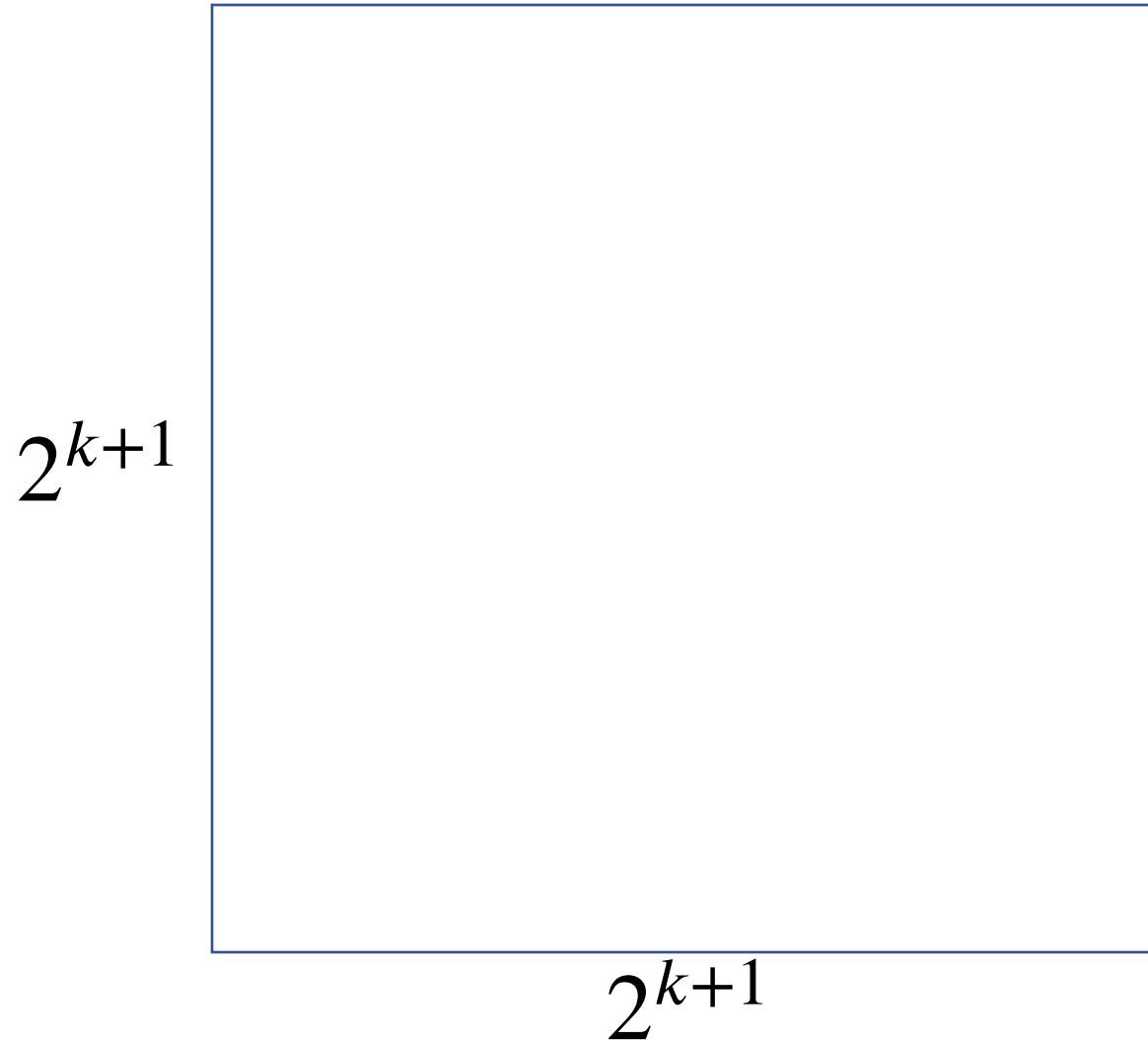


Triominoes: Inductive Case

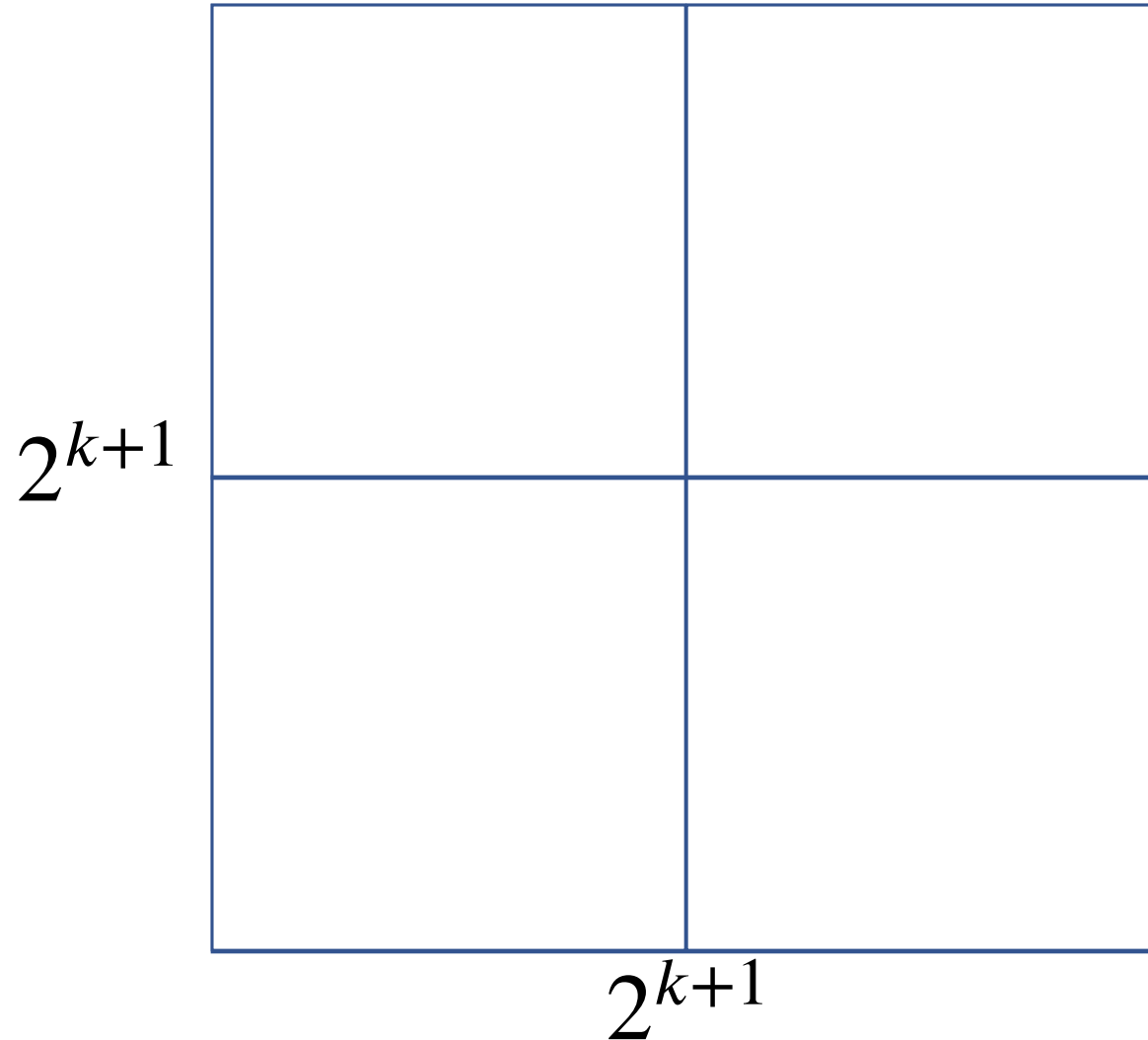

$$2^{k+1}$$

$$2^{k+1}$$

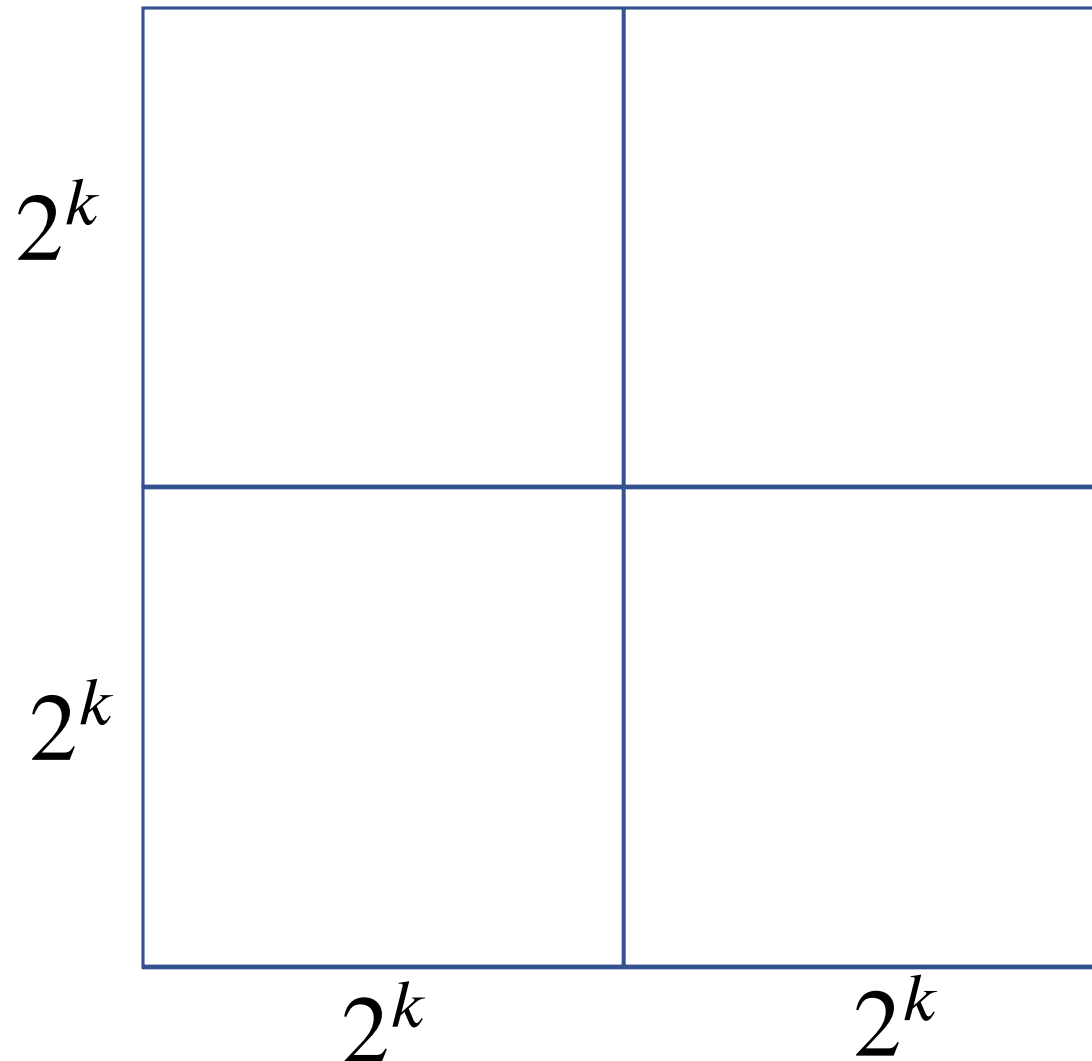
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



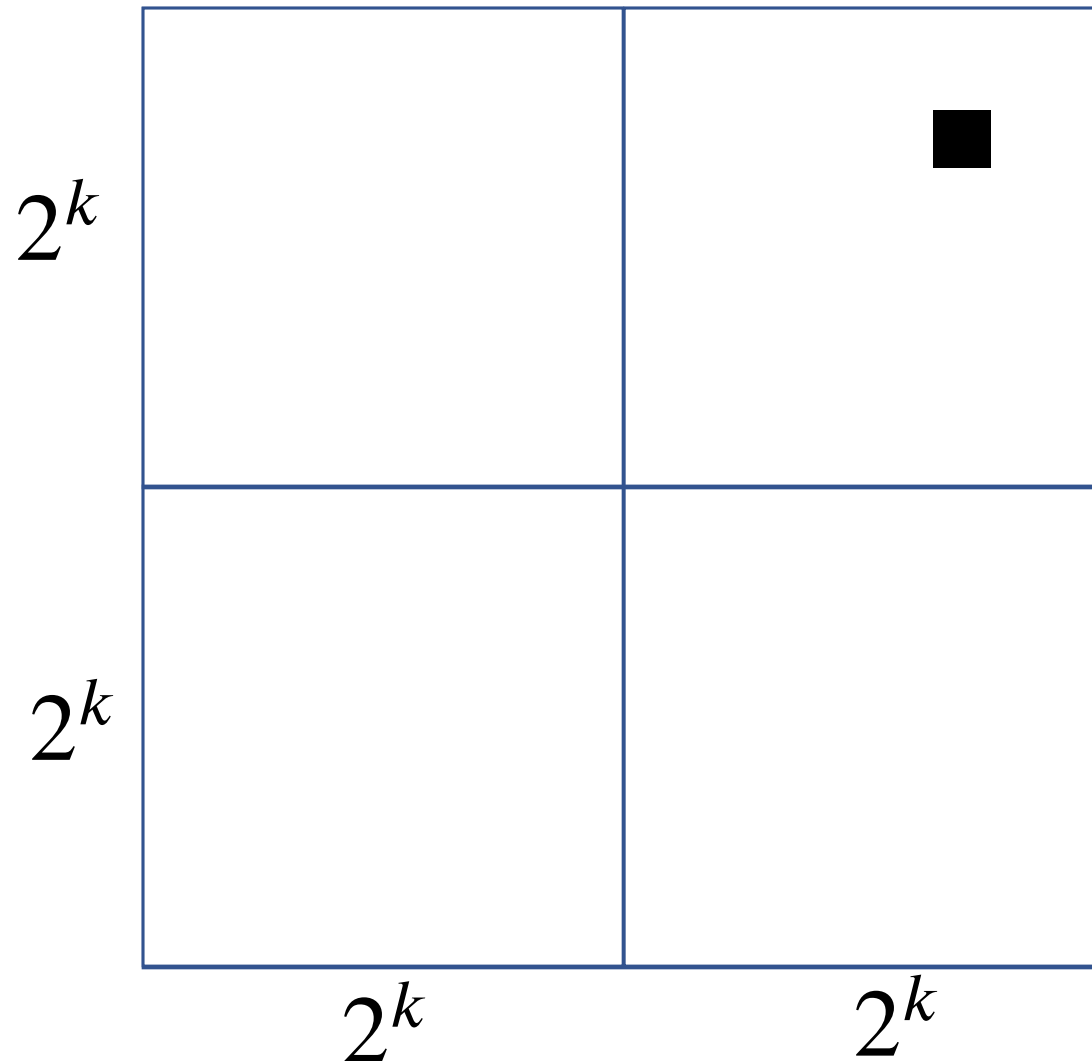
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



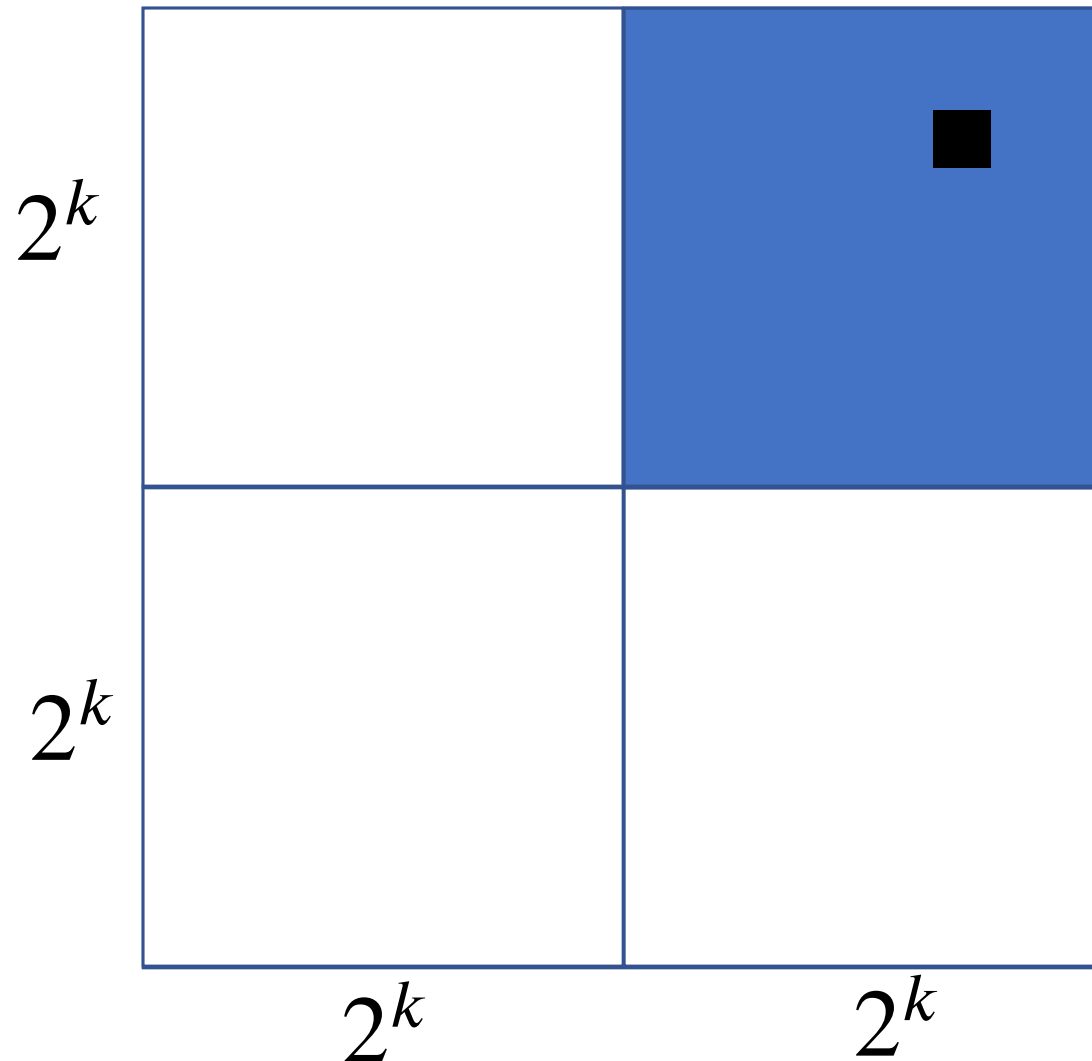
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



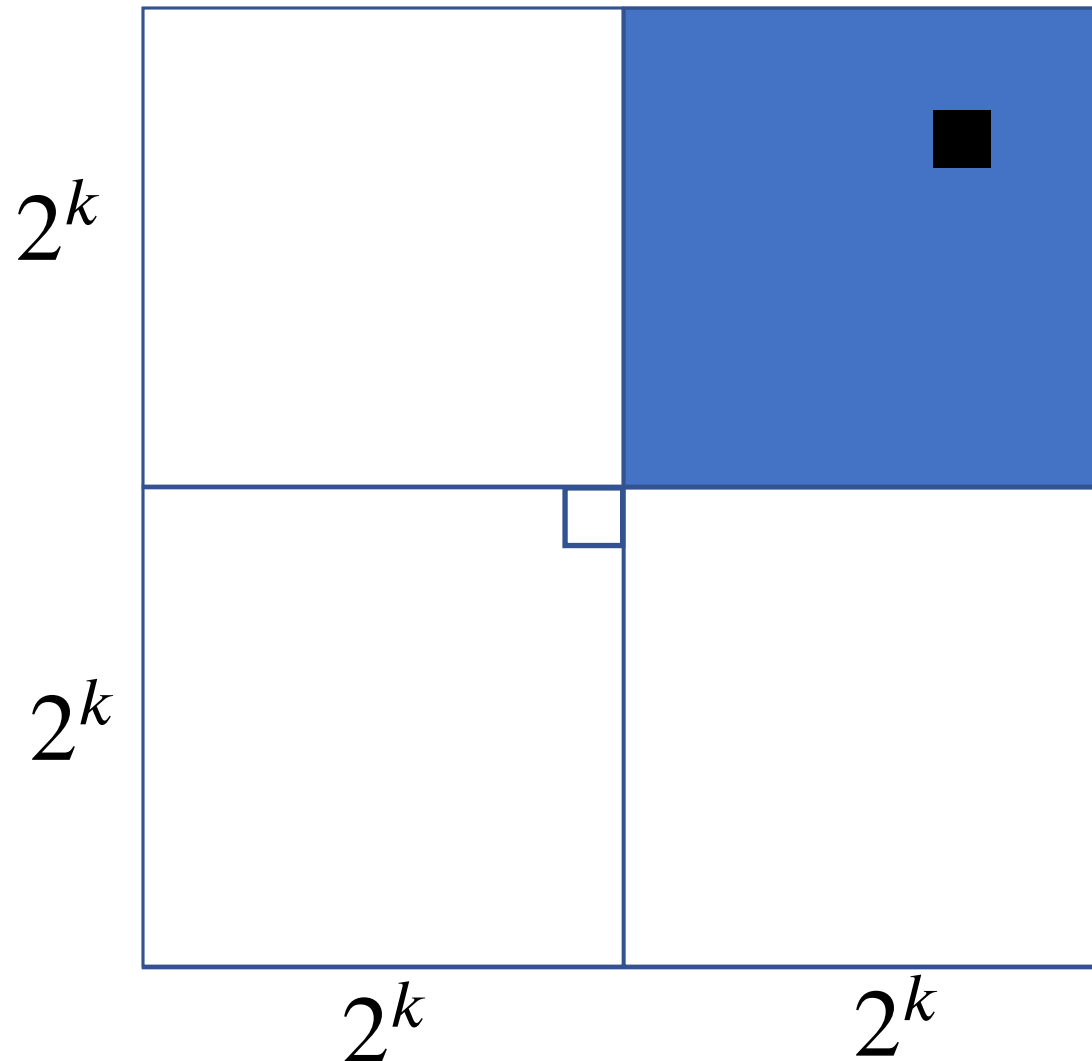
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



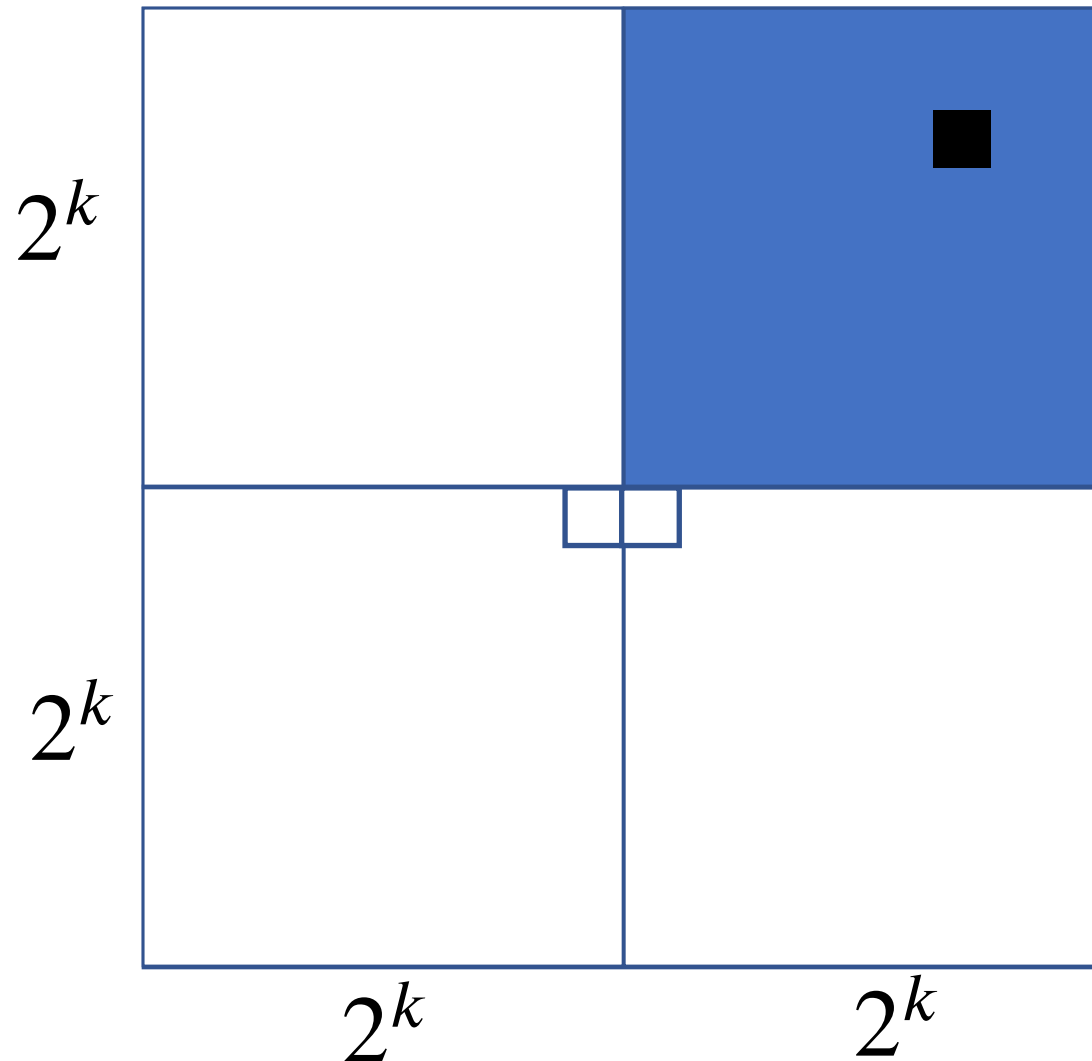
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



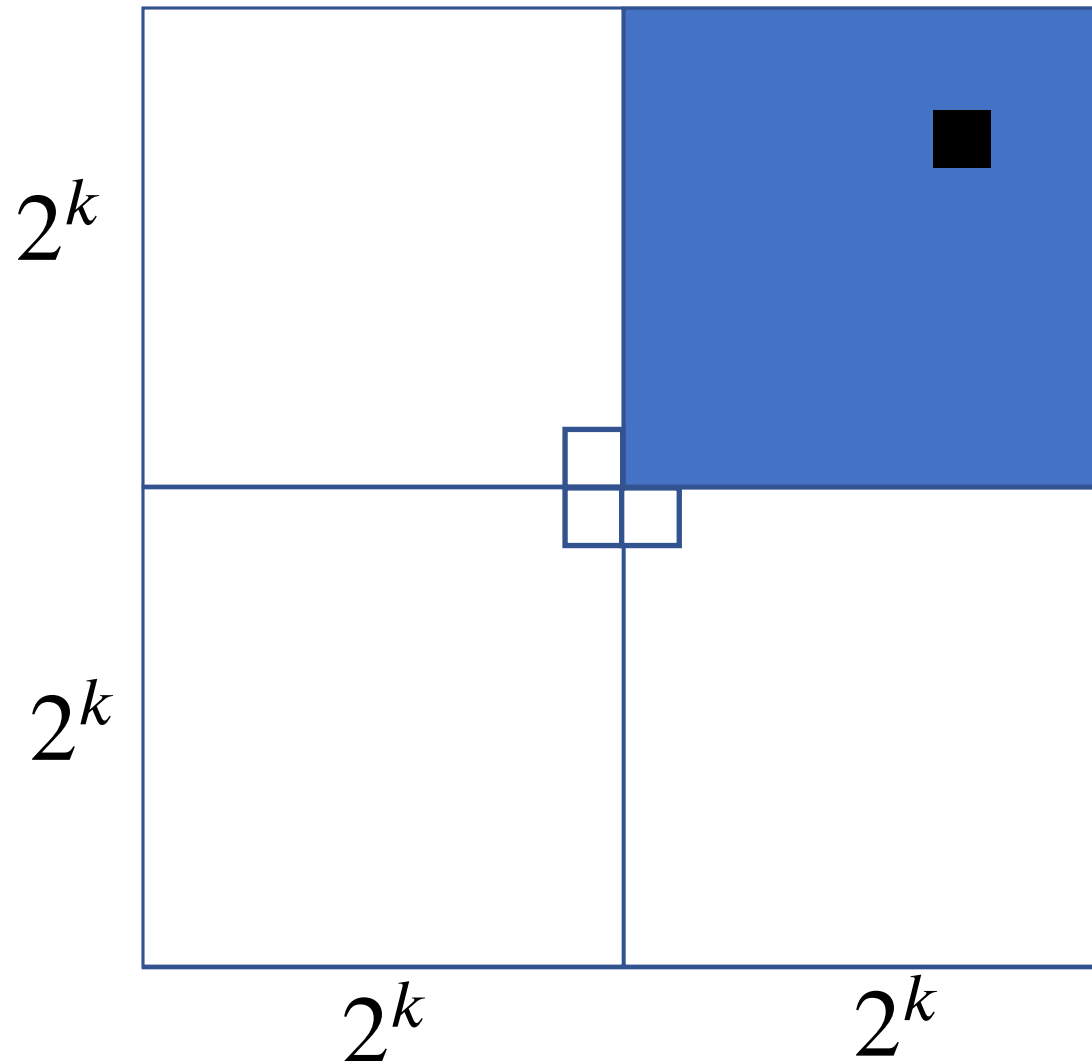
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



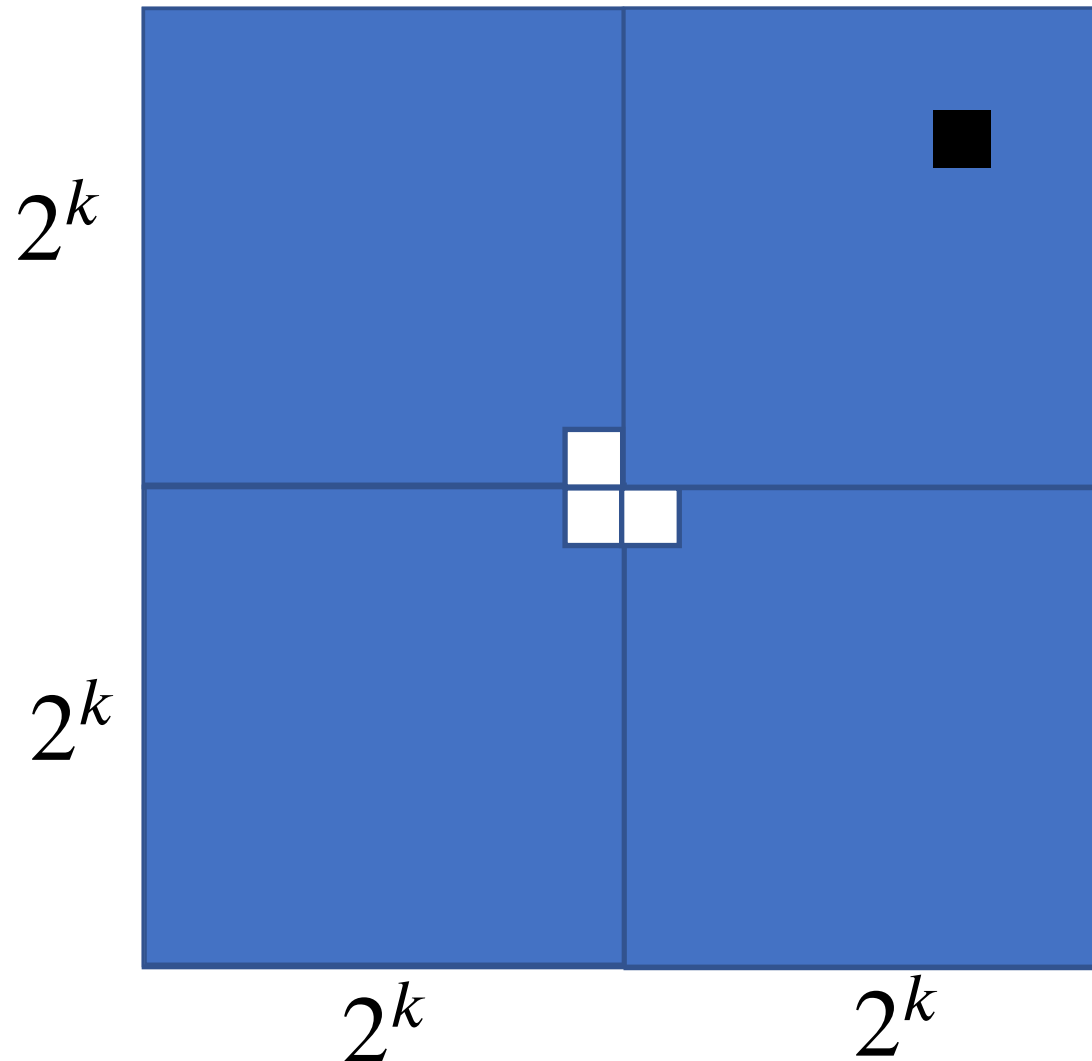
Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes



Inductive hypothesis: Any checkerboard of size $2^k \times 2^k$ with a hole in it can be tiled with triominoes

